Local Evidence Is Not Completion: Evidence-Condition Geometry for Bounded Completion-Audit Workflows

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Abstract

Completion-audit workflows often stop before the remaining workload is known. Their stop signals–no-new findings, agent agreement, stable summaries, or self-reported completion–can be locally meaningful while still being unsafe as scope-wide completion certificates. The missing question is: under what evidence condition was the stop signal produced, and does that condition match the claimed scope? We study this certificate mismatch in bounded source-route audit workflows. A source-route stratum records both where the workflow searched and which audit route it applied. We use the resulting exposure geometry to build an evidence-condition controller with four modules: source-route exposure estimation, an eligibility gate, residual-evidence repair over weak plausible gaps, and a final SAFE/CONTINUE/ABSTAIN decision rule. Across four bounded completion-audit task families, homogeneous route reuse creates false certification risk; source-route granularity detects this risk where source-only control does not; broad eligibility alone is insufficient on the `urllib3` boundary; and the full controller avoids unsafe stops without becoming a never-stop rule. The result is a bounded diagnostic and control principle: completion certificates should be judged by the evidence condition that produced them, not by stop signals alone.

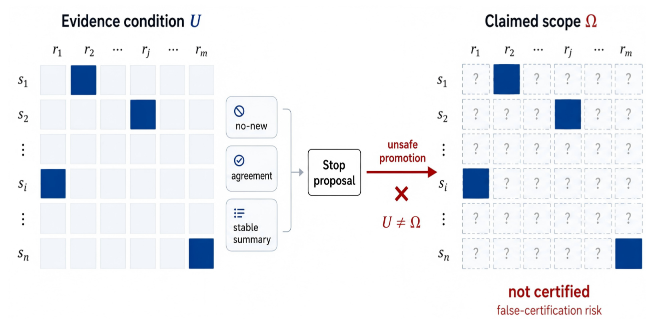

Figure 1: Certificate mismatch from local stop evidence. A workflow may observe no-new findings, agreement, or a stable summary under evidence condition U. Promoting that stop proposal to a completion claim over Ω is unsafe when $U \neq \Omega$; the unobserved source-route strata are not certified.

1 Introduction

Completion-audit workflows must eventually stop, even when the remaining workload is unknown. A local stop signal–no new findings, agreement, or a stable summary–can be true under the evidence condition that produced it while still failing as a certificate for the declared task scope. We call this *certificate mismatch*. In a code audit, for example, exhausting timeout evidence in a file does not certify TLS, retry, exception, or cleanup evidence in that same file.

We model this mismatch with *evidence-condition geometry*. A source-route stratum records both where a workflow searched and which audit route it applied. The exposure distribution over these strata is a runtime certificate diagnostic: it tells us whether a proposed stop is supported by broad source-route evidence or by a localized condition. The diagnostic does not know the hidden workload and does not prove open-world completion.

We build a controller around this diagnostic. It estimates source-route exposure, checks eligibility, repairs weak plausible strata, and returns SAFE/CONTINUE/ABSTAIN. A generic verifier gate can avoid unsafe SAFE decisions by failing closed, but it gives no source-route diagnosis or repair target. Our controller aims for a stronger operational property: safe decisions when the condition is defensible, actionable CONTINUE decisions when residual evidence is found, and explicit ABSTAIN when no certificate can be formed. The contribution is therefore not a new retrieval policy alone; it is a certificate interface for deciding whether a stop proposal is safe, actionable, or still unsupported.

Experiments on two generated audits and two frozen real-repository audits test this safety–actionability–cost tradeoff. Homogeneous route reuse creates false certification risk; source-route geometry exposes risks hidden by source-only coverage; the `urllib3` boundary shows that eligibility is not proof; and the full controller avoids unsafe SAFE decisions while turning residual-positive unsafe states into actionable CONTINUE rather than merely abstaining. The paper makes three claims. First, stop evidence has a condition, and local conditions cannot certify broader source-route scopes. Second, source-route geometry is a useful stop-time warning signal because it separates file coverage from route coverage. Third, a repair-aware controller is more useful than a generic fail-closed verifier because it preserves safety while producing the next audit target when residual evidence appears.

The claim is bounded: completion certificates should be judged by the evidence condition that produced them, not by stop signals alone.

2 Related Work

High-recall review, total recall, active search, and review stopping rules ask how to retrieve more positives, estimate remaining mass, or decide whether a review has saturated under budget [Grossman and Cormack, 2011, Cormack and Grossman, 2014, 2015, 2016, Grossman et al., 2016, Roegiest et al., 2015, Voorhees, 2002, Wallace et al., 2010, Lewis and Gale, 1994, Settles, 2009, Garnett et al., 2012, Attenberg et al., 2015, Good, 1953, Chao, 1984, Chao et al., 2014]. Our controller is not given oracle totals, recall, hidden missing mass, or unseen true-item counts. It asks a different stop-time question: whether the runtime evidence condition matches the declared source-route scope of the completion claim.

Tool-using, web, code, and multi-agent systems allocate work across tools, agents, memories, searches, and intermediate states [Yao et al., 2023, Schick et al., 2023, Nakano et al., 2021, Qin et al., 2024, Wu et al., 2023, Park et al., 2023, Wang et al., 2024, Deng et al., 2024, Zhou et al., 2024, Liu et al., 2024, Yao et al., 2022, Jimenez et al., 2024, Shinn et al., 2024, Madaan et al., 2023]. Benchmarks and factuality systems evaluate final task success, answer support, or post-hoc correctness [Hendrycks et al., 2021a, Liang et al., 2022, Srivastava et al., 2023, Lin et al., 2022, Thorne et al., 2018, Gao et al., 2023, Min et al., 2023, Schuster et al., 2021, Gehrmann et al., 2021]. Verification and admission-control pipelines can fail closed when a checker reports unresolved warnings. We use such a verifier-gate as a baseline. The distinction is the certificate object: evidence-condition control diagnoses where the stop certificate is weak and uses repair to produce actionable CONTINUE decisions, not only reject or abstain. This connects to reliability and oversight concerns [Dietterich, 2000, Cobbe et al., 2021, Wang et al., 2023, Amodei et al., 2016, Christiano et al., 2017, Irving et al., 2018, Leike et al., 2018, Hendrycks et al., 2021b], but targets a narrower operational failure: locally conditioned evidence being promoted to a global completion certificate.

3 Problem Setting

We study bounded completion-audit workflows with unknown residual workload and declared source-route scope. A workflow receives a task, allocates work across agents or tools, gathers evidence under a finite budget, and must decide whether the task is complete within that declared scope. Let

$$W = (T, S, R, A, B),$$

where T is the task, S is a set of sources, R is a set of routes, A is the set of agents or tools, and B is a finite budget. A source is a bounded evidence region, such as a file, module, document, source family, or repository component. A route is an audit or search lens applied to a source.

At runtime, the workflow produces evidence events

$$e_t = (a_t, x_t, r_t, u_t, o_t, c_t),$$

where a_t is the agent or tool, $x_t \in S$ is the source, $r_t \in R$ is the route, u_t is the action, o_t is the observation, and c_t is the cost. Every evidence event is produced under a source-route condition. A source-route stratum is $s = (x, r)$, and the intended evidence-condition space is

$$\Omega = S \times R.$$

The source-route space does not require the workflow to know all true missing items in advance. It requires the intended audit scope to be defined: which sources and routes are relevant to the completion claim. This assumption is weaker than knowing the answer set and stronger than an unbounded open-world search. It matches many practical audits: a system may know which repositories, documents, modules, or evidence sources are in scope, and it may know which audit routes are relevant, while still not knowing how many true findings remain. The controller therefore evaluates the condition under which evidence was gathered, not the unknown total workload itself.

A stop claim is a workflow assertion

$$C_t : \quad T \text{ is complete over the intended workload scope.}$$

A false certification occurs when the workflow accepts SAFE for a stop claim that is not supported by the evaluation oracle:

$$\textsc{Safe}(C_t) = \text{true}, \qquad \text{complete}(T) = \text{false}.$$

In item-discovery subclasses, this is operationalized as accepting SAFE while bounded-oracle recall is below the completion threshold. In broader completion audits, it means accepting a global completion claim while relevant support, contradiction, or unresolved evidence remains outside the evidence condition. The oracle is used only for evaluation. It is not available to route assignment, repair scoring, eligibility checks, or stop decisions.

4 Evidence-Condition Geometry

Let $v_t(s)$ be the runtime-visible exposure count for source-route stratum s up to time t. Exposure may count visits, scans, route-specific audits, tool calls, or other logged search actions. Define

$$p_{\text{exp}}(t, s) = \frac{v_t(s)}{\sum_{s' \in \Omega} v_t(s')}.$$

This distribution is a point on the source-route coverage simplex. It describes where and how completion evidence was produced.

We use two runtime summaries:

$$\text{support}(t) = \frac{|\{s : v_t(s) > 0\}|}{|\Omega|},$$

$$G_{\text{exp}}(t) = \text{Gini}(\{v_t(s) : s \in \Omega\}).$$

These summaries are operational diagnostics, not universal laws. They summarize whether the evidence condition is localized or broad enough to make a completion certificate eligible for consideration.

Proposition 1 (Local no-new evidence is not a global certificate)**.** *Let $U \subset \Omega$ be a strict subset of the intended source-route space. Suppose all evidence used to support a stop claim C_t is produced inside U, and suppose the workflow observes no-new evidence only for actions inside U. Then the no-new evidence can support local exhaustion over U, but it cannot by itself support global completion over Ω.*

Proof sketch. No-new evidence is conditioned on the source-route actions that were executed. If no events are produced in $\Omega \setminus U$, the workflow has not observed the absence of residual evidence in those unexposed strata. A world in which U is exhausted and $\Omega \setminus U$ contains residual evidence is observationally indistinguishable from a globally complete world under the workflow's local evidence log. Therefore local no-new evidence rules out additional findings only under the searched condition U; without additional assumptions about unsearched strata, it cannot rule out residual evidence in $\Omega \setminus U$.

Source-only geometry is too coarse: auditing a file for timeout behavior does not certify that TLS, retry, exception, or cleanup behavior has been audited in that file. Source-route-action geometry is more detailed, but it is sensitive to logging conventions and current experiments do not show a stable advantage over source-route. Source-route is the default because it captures both where and how evidence was produced while remaining interpretable and runtime-computable. The geometry is thus a certificate diagnostic rather than a semantic theory of all possible evidence. It asks whether the current log covers the intended source-route space broadly enough for the proposed stop claim; it does not assert that all routes are equally important or that support/Gini thresholds transfer unchanged across domains.

5 Method

The controller has four modules, shown in Algorithm 1. *Exposure estimation* maps the runtime evidence log to source-route counts $v_t(s)$, support $\mathrm{support}(t)$, and Gini concentration $G_{\exp}(t)$. *Eligibility* checks whether the stop condition is broad enough to consider a certificate. *Residual repair* probes weak plausible strata using only runtime-visible signals. The final rule returns SAFE only when the evidence condition is eligible and repair is nonproductive; residual evidence yields actionable CONTINUE, and unresolved ineligibility yields ABSTAIN.

The repair instance used in experiments ranks strata by

$$\mathrm{priority}(s) = \mathrm{under\_exposure}(s) \cdot \mathrm{runtime\_potential}(s).$$

The potential term may use source text, route names, route match counts, source size, or ledger-visible notes. It may not use oracle labels, oracle totals, post-hoc recall, hidden missing mass, or undiscovered true-item counts. Residual-potential is a mechanism-aligned probe, not an optimal active-search claim. Thus the method's value is stop-time certificate diagnosis plus targeted repair, not universal completion proof.

Algorithm 1 Evidence-Condition Controller

Require: log E_t, space Ω, claim C_t, budget B, thresholds τ_s, τ_g, repair rule $\mathcal{R}$
Ensure: decision in $\{\text{SAFE}, \text{CONTINUE}, \text{ABSTAIN}\}$

Compute $v_t(s)$, $p_{\exp}(t, s)$, $\mathrm{support}(t)$, and $G_{\exp}(t)$
$eligible \leftarrow [\mathrm{support}(t) \geq \tau_s \wedge G_{\exp}(t) \leq \tau_g]$
if not $eligible$ and $B = 0$ **then**
  return ABSTAIN
end if
if not $eligible$ or weak plausible gaps remain **then**
  Audit $Q = \mathcal{R}(E_t, \Omega, C_t)$ within B
  if residual evidence appears **then**
    return CONTINUE
  end if
  Recompute $eligible$
end if
if $eligible$ and no residual evidence appeared **then**
  return SAFE
end if
return ABSTAIN

Proposition 2 (Controller certificate semantics)**.** *For a declared source-route scope Ω, thresholds τ_s, τ_g, and a fixed repair rule $\mathcal{R}$ that uses only runtime-visible signals, Algorithm 1 returns* SAFE *only if (i) the exposure condition satisfies the eligibility gate and (ii) repair or audit reveals no residual evidence within the allocated repair budget. If residual evidence is found, the returned decision is* CONTINUE*; if eligibility cannot be established under budget, the returned decision is* ABSTAIN.

The proof sketch is in the supplement; the proposition is about the controller's declared semantics, not about absence of all unobserved evidence.

6 Experiments

We evaluate whether a proposed stop is *safe, actionable, and cost-aware* under a declared source-route scope. Each run fixes a workflow trajectory before scoring: the decision rule observes only runtime-visible evidence, optional repair scans the fixed budget, and oracle labels are revealed only afterward to score false certification. The experiments therefore test completion control, not oracle-assisted discovery. Three invariants make the comparison closed. First, every policy is evaluated on the same proposed stop states, so gains do not come from changing the trajectory. Second, oracle labels are used only after a decision is fixed, so the controller cannot read oracle totals, recall, or undiscovered true items. Third, CONTINUE and ABSTAIN are separated: CONTINUE is useful only when repair finds residual evidence and yields a next audit target, whereas ABSTAIN is a conservative refusal to certify.

Tasks and states. We use four bounded audit families: two generated hidden-oracle tasks (`policy-docset`, 24

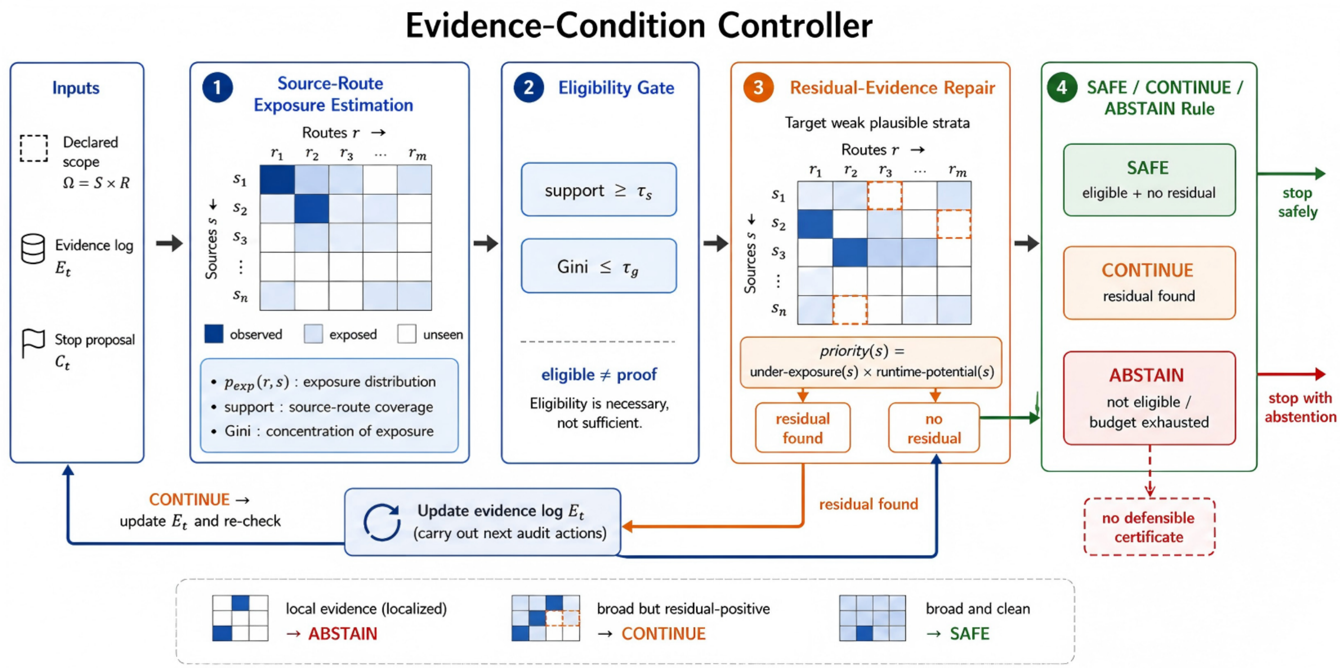

Figure 2: Evidence-condition controller. The controller maps a runtime evidence log, declared source-route scope, and stop proposal to an exposure distribution over the intended source-route space. The support/Gini check is an eligibility test, not a completion proof. If the condition is localized or weak plausible gaps remain, the controller repairs or continues. It returns SAFE only when the evidence condition is broad and repair or audit reveals no residual evidence; otherwise it returns CONTINUE or ABSTAIN.

clauses over four policy routes; `code-repo`, 20 risks over three code routes) and two frozen external repository audits (`requests`, 298 pattern-defined items over TLS/timeout/exception/ compatibility; `urllib3`, 699 items over timeout/retry/TLS/exception/ cleanup). Homogeneous-route stops reuse one route across sources; route- partitioned stops spread routes across the same sources; `urllib3` also has an extended audit covering all five routes. External oracles are frozen, pattern-defined mechanisms for scoring, not human semantic completion labels. The generated tasks make the hidden oracle exact; the external tasks make the audit surface realistic but pattern-defined. The point is mechanism validation across task types, not a claim that regex-style repository oracles replace expert semantic annotation.

Ablations. We report five ablations. The *granularity ablation* compares source-only and source-route exposure. The *proof-separation ablation* compares eligibility-only with the full repair-aware controller. The *decision-rule ablation* compares Naive stop, Source-only, Verifier-gate, Eligibility-only, and Full. The generic Verifier-gate ignores source-route geometry and returns SAFE only when no unresolved or residual warning remains; otherwise it fails closed to ABSTAIN. The *repair-target ablation* compares Random, High-potential, and Residual-potential targets under the full rule. The *threshold/budget sensitivity* sweep varies τ_s, τ_g and repair budget to expose safety-cost choices.

Reproducibility and leakage control. Generated tasks use repair budget 4; `requests` and `urllib3` use 200 seeded validations with budgets 4 and 5. The operational thresholds are $\tau_s = 0.75$, $\tau_g = 0.70$; the 0.90 recall threshold is post-hoc only. Configs specify seeds, routes, thresholds, budgets, oracle paths, and output paths. Runtime rules may use evidence logs, source text, route names, source sizes, and ledger-visible notes; they may not use oracle labels, oracle totals, post-hoc recall, or undiscovered true-item counts. Full tables, run commands, and exported CSV/JSON manifests are in the supplement. Metrics are tied to state type. FCR is SAFE-on-unsafe divided by seeded unsafe states; safe coverage is SAFE-on-complete divided by seeded complete states; actionability is CONTINUE-on-unsafe, not merely non-SAFE; and repair cost counts post-stop scanned lines plus extraction events. This denominator discipline is important because the single `urllib3` boundary case and the seeded controller counts answer different questions.

7 Results

The main result is a safety-usefulness-cost tradeoff. The full controller avoids unsafe SAFE decisions like a conservative verifier gate, but unlike a fail-closed verifier it returns actionable CONTINUE when residual evidence points to the next audit target. We organize the evidence as a diagnostic chain: local stop signals → source-only illusion → source-route mismatch → eligibility is not proof → residual repair closes the loop.

Local stop signals are not certificates. Figure 3 begins with the failure mode. Across all four tasks, homogeneous route reuse produces localized evidence and would falsely certify completion if accepted as SAFE. Controlled sweeps in the supplement show the same qualitative pattern under localized exposure. This motivates source-route granularity rather than accepting unconditioned stop signals. This validates the first link in the diagnostic chain: local stop evidence can be real while still being the wrong certificate for a scope-wide claim.

Source-only hides route mismatch. The middle panel of Figure 3 is the source-only vs. source-route ablation. Homogeneous runs touch all files, so source-only support looks complete; the same states have localized source-route support and sub-threshold recall. Source-only exposure cannot detect that timeout evidence does not certify TLS, retry, exception, or cleanup routes in the same file. In the four task families, source-only support is 1.0 at the homogeneous stop, while source-route support ranges only 0.20–0.33 and bounded-oracle recall ranges 0.104–0.708. This validates the second link: source coverage alone can hide a route-conditioned mismatch.

Source-route geometry exposes mismatch, not proof. The source-route view records both where evidence was gathered and which route produced it. In homogeneous conditions, this geometry stays concentrated even when source-only support is full; in route-partitioned and extended audits, support broadens and concentration falls. The diagnostic value is therefore not that source-route metrics prove completion, but that they expose when the stop claim is conditioned on too narrow an evidence route. This validates the third link: source-route geometry turns hidden route mismatch into an observable stop-time warning signal.

Eligibility is only a precondition. Route-partitioned evidence improves eligibility, but it is not proof. In `urllib3`, route-partitioned exposure covers 24/30 strata and passes the gate, yet recall remains below threshold and repair finds residual evidence; the controller returns CONTINUE, not SAFE. This is a single fixed boundary case: support is 0.800, Gini is 0.647, recall is 0.835, and the post-hoc oracle still contains 115 missed items. Its denominator is one eligible-but-residual-positive `urllib3` state, whereas the seeded controller counts in the supplement use unsafe-state and complete-state denominators. The two denominators must not be mixed. The extended audit reaches support and recall 1.000. This validates the fourth link: eligibility permits a completion claim to be considered, but does not certify it.

Full control is safe and actionable. Figure 4 evaluates decision rules on the same seeded unsafe repair states and seeded complete-state perturbations. Naive and source-only rules certify unsafe local stops. Verifier-gate avoids unsafe SAFE decisions by failing closed to ABSTAIN whenever warnings remain, but it does not identify the under-covered source-route condition or a repair target. The full controller keeps FCR at zero in the seeded external validation, preserves safe coverage on 1200 seeded complete states, and converts all 400 productive seeded unsafe repair states into CONTINUE rather than opaque ABSTAIN. The contrast with verifier-gate is sharp: both have FCR 0, but verifier-gate has unsafe-state ABSTAIN rate 1.0, while the full controller has unsafe-state CONTINUE rate 1.0. This is the practical difference: ABSTAIN says "do not certify," while CONTINUE says where the audit should continue. This validates the safety and usefulness part of the chain.

Repair cost is an explicit tradeoff. Figure 4 separates repair target variants from decision variants. Residual-potential exposes residual evidence and has higher mean gain than random repair, but it is not proven optimal; high-potential is competitive in cost-normalized terms. Repair is therefore a bounded probe for remaining evidence, not a claim of optimal active search. The supplement includes the controller count tables, per-task breakdowns, source-only and scalar negative controls, and the full threshold/budget sweep. The sensitivity result is stable: changing τ_s, τ_g and budget changes the SAFE/CONTINUE/ABSTAIN mixture and cost, but does not create false certifications in the external validation. In the seeded unsafe repair states, residual-potential finds 253.0 new oracle items at mean cost 4275.5; random repair finds 69.0 at lower mean cost 3481.1; and high-potential finds 226.0 at mean cost 3824.0. Thus the method buys actionability with explicit extra audit cost, not with a free certificate. This validates the cost part of the chain: residual repair turns a stop into a bounded decision with an auditable next action.

Sanity checks. A small deterministic claim-verification pilot uses human-style routes (support, contradiction, exception-path, config-default, and scope-boundary) only as an interface sanity check, not as a main benchmark. It shows the same pattern: Verifier-gate can avoid unsafe SAFE by abstaining, while the full controller uses source-route diagnosis and residual repair to avoid false certification without giving up safe coverage in this pilot. Additional oracle sanity, leakage, safe-state, and sensitivity checks are in the supplement. This validates the final link: residual repair converts an eligible stop into a closed SAFE/CONTINUE/ABSTAIN decision rather than a bare metric threshold.

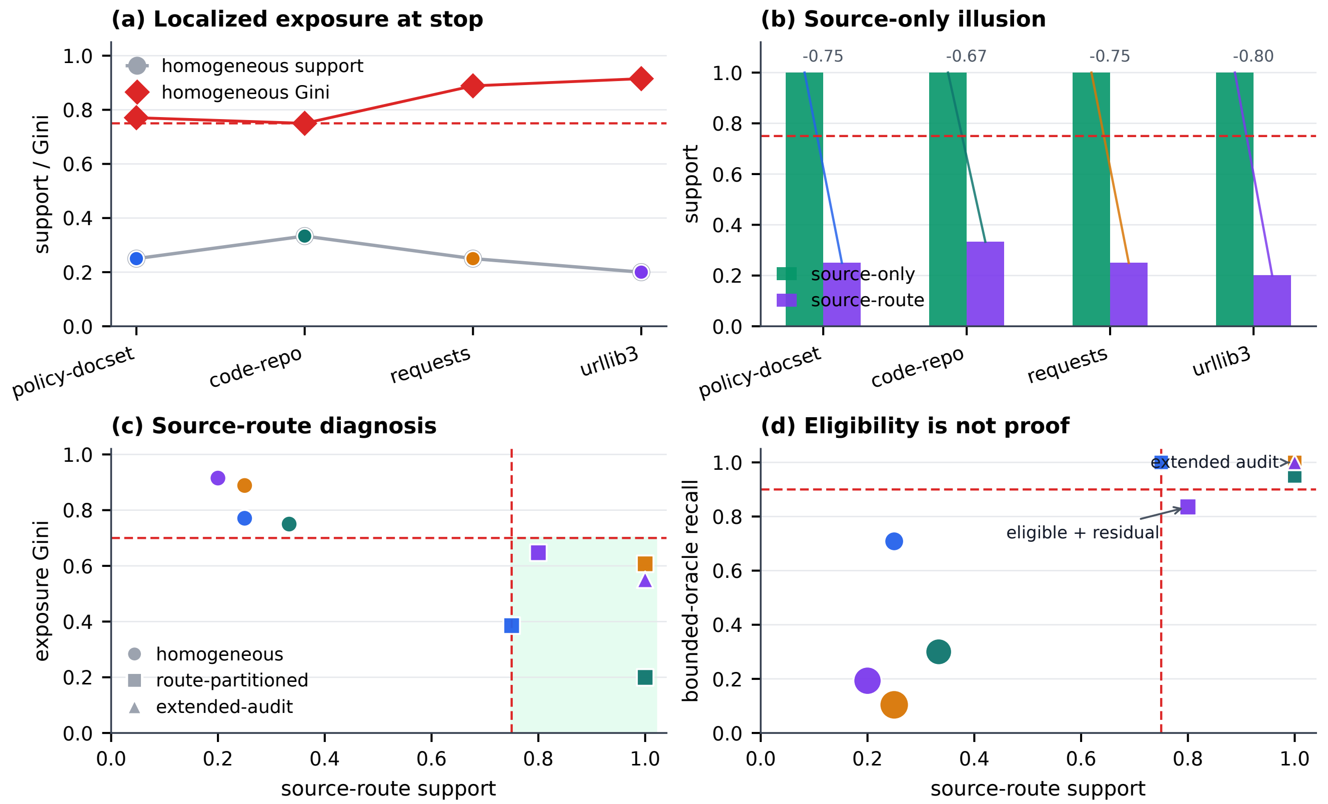

Figure 3: Diagnostic chain and granularity ablation. Local homogeneous stops have low source-route support and high concentration; source-only support is misleadingly complete; source-route support/Gini reveal the mismatch; and the `urllib3` route-partitioned state is eligible but residual-positive, with recall 0.835 below the post-hoc 0.90 threshold until extended audit reaches 1.000.

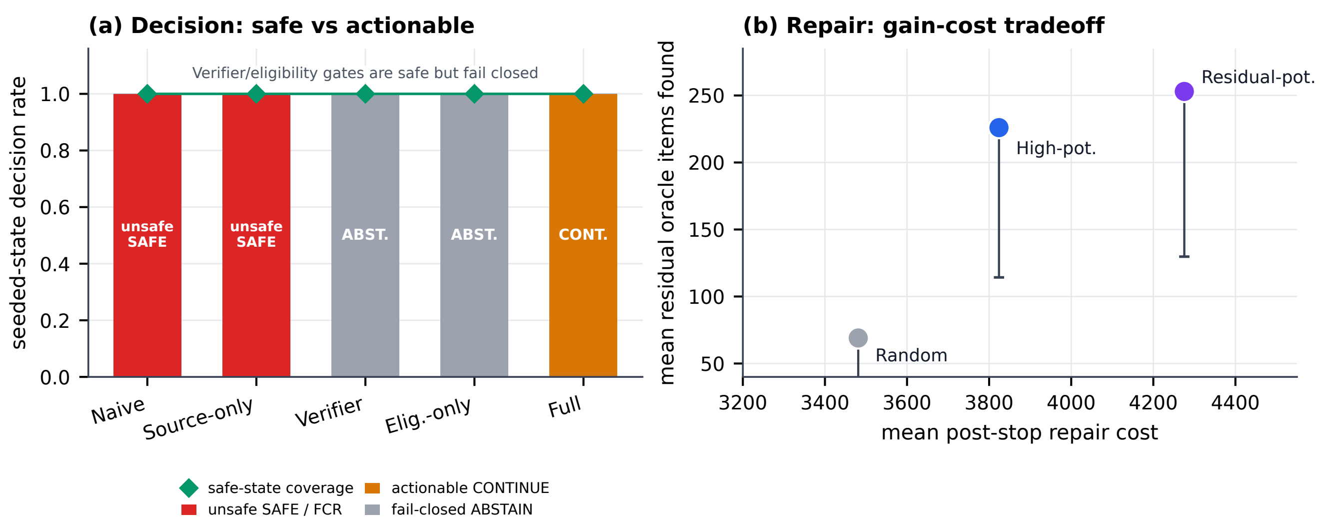

Figure 4: Controller evaluation dashboard for safe, actionable, and cost-aware stopping. The decision panel compares safety and usefulness: verifier-gate is safe but fail-closed, while the full controller is safe and actionable on productive unsafe states. The repair panel compares target rules and budget choices, showing residual-potential as a high-yield but non-optimal repair heuristic.

8 Discussion and Limitations

The positive result is bounded but operationally useful. A SAFE decision means that the declared source-route condition is eligible and that budgeted repair did not reveal residual evidence; it is not a universal guarantee that no further facts exist. In practice, support and Gini act as stop-time warning signals, threshold and budget sweeps set conservatism before deployment, and residual repair turns a weak certificate into a next audit target. The SAFE/CONTINUE/ABSTAIN interface is useful precisely because it distinguishes certified stops, productive continuation, and conservative refusal.

The limitations are real. External repository oracles are pattern-defined over frozen snapshots, not human semantic gold labels. They stress certificate logic at realistic scale, but they do not prove that an expert would agree with every completion boundary. Residual-potential repair is also a heuristic, not an optimal active-search policy. The comparison with high-potential repair is therefore intentionally modest: residual-potential is mechanism-aligned and high-yield in these states, but it is not a theorem about the best next action. The claim-verification pilot is only a small interface sanity check.

These limits sharpen the main point. Local stop signals are not completion certificates. In bounded audit workflows, the defensible question is whether the evidence condition supporting the stop matches the declared scope and whether residual repair can convert uncertainty into an actionable next audit step.

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